Piecewise Functions: Answer Key

Precalculus · Use this to check your own work on the Piecewise Functions notes and homework. Answers only, so redo any problem you missed before looking at the answer again.

Big idea: a piecewise function is made up of 2 or more functions, each with its own domain.

Notes: Evaluating Piecewise Functions

Problem	Answer
1a. $f(-5)$	20
1b. $f(2)$	4
1c. $f(7)$	$2\sqrt{5}$
2a. $g(-3)$	8
2b. $g(-8)$	-249
2c. $g(3.1)$	-1

Notes: Graphing Piecewise Functions

Problem	Domain	Range	Discontinuities
3	$\mathbb{R}$	$\{y \mid y \geq -5\}$	$x = 0$, jump
4	$\mathbb{R}$	$\mathbb{R}$	$x = -3$, jump
5	$\{x \mid x \neq -3\}$	$\mathbb{R}$	$x = -3$, infinite $x = 1$, jump
6	$\mathbb{R}$	$\mathbb{R}$	$x = -4$, jump $x = 2$, jump
7	$\mathbb{R}$	$\{y \mid y \leq 7\}$	$x = -1$, jump $x = 2$, jump

Homework 8: Evaluating

Problem	Answer
1a. f(−4)	−65
1b. f(−2)	−1
2a. g(−7)	−42
2b. g(4)	$3\sqrt{3}$
3a. f(−1.2)	−4
3b. f(2)	7
4a. h(−5)	50
4b. h(8/15)	39/5 (which is 7.8)

Homework 8: Graphing

Problem	Domain	Range	Discontinuities
5	$\mathbb{R}$	$\{y \mid y \le 7\}$	x = 3, jump
6	$\{x \mid x \ne 1\}$	$\mathbb{R}$	x = 1, infinite x = 3, jump
7	$\mathbb{R}$	$\{y \mid y \ge -7\}$	x = −2, jump x = 2, jump
8	$\{x \mid x \ne -4\}$	$\{y \mid -7 \le y < 1 \text{ or } y > 1\}$	x = −4, removable (a hole) x = 1, jump
9	$\mathbb{R}$	$\{y \mid y \ge -3\}$	x = −5, jump x = −2, jump x = −1, jump, and at every integer after that

Checking your graphs:

- Graph each piece only on its own domain, and stop at the boundary value.
- Use a closed dot where the inequality includes the boundary (≤ or ≥) and an open dot where it does not (< or >).
- A **jump** means the two pieces do not meet at the boundary. A **removable** discontinuity (hole) means the pieces head to the same height but the point is missing. An **infinite** discontinuity is a vertical asymptote.
- To find the range, look at the height covered by each piece, then combine them. Pieces can overlap.